

Autonomous Fire Fighting Robot

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Introduction:

Robotics is widely used in hazardous environments to improve safety. Firefighting robots can autonomously detect and extinguish fires, reducing risks to humans. Arduino-based robots make such systems cost-effective.

Problem Statement:

Fires pose serious risks to life and property. This project aims to build an Arduino-based robot that detects fire, avoids obstacles, and extinguishes it safely.

Importance:

- Enhances safety by reducing human risk.
- Low-cost and easy to build.
- Useful in homes, hostels, and small industries.
- Provides practical learning in robotics and automation

Objectives:

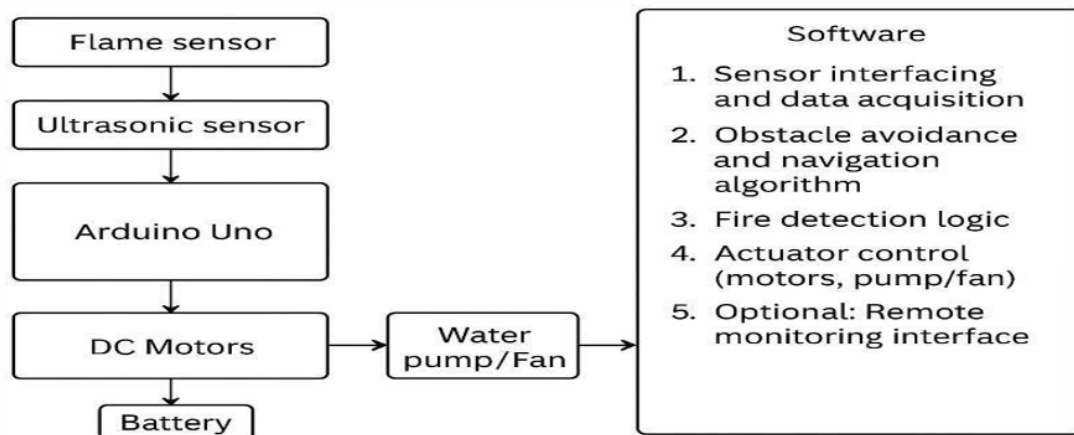
- Develop autonomous navigation to move safely in obstacle-filled environments.
- Detect fire accurately using flame or IR sensors.
- Activate fire-extinguishing mechanism automatically when fire is detected.
- Improve obstacle avoidance using ultrasonic sensors.
- Create a low-cost, compact, and reliable robotic system.
- Enhance safety by reducing human involvement in fire-prone areas.

Literature Review :

- Existing Robotic Systems
 - Industrial firefighting robots use heavy machinery and advanced sensors.
 - Some small robots are remote-controlled for fire monitoring.
 - Most rely on expensive hardware or human supervision.

- Limitations of Current Approaches
 - High cost and complex maintenance.
 - Limited autonomy; often require human control.
 - Bulky, not suitable for small or cluttered spaces.
 - Sensors may give false alarms in sunlight or reflective areas.
- Opportunity for Improvement
 - Build a compact, low-cost, fully autonomous robot.
 - Improve navigation and fire detection in small or cluttered spaces.
 - Use affordable microcontrollers like Arduino for easy implementation.
 - Enhance safety by reducing human intervention.

System Architecture:



Hardware:

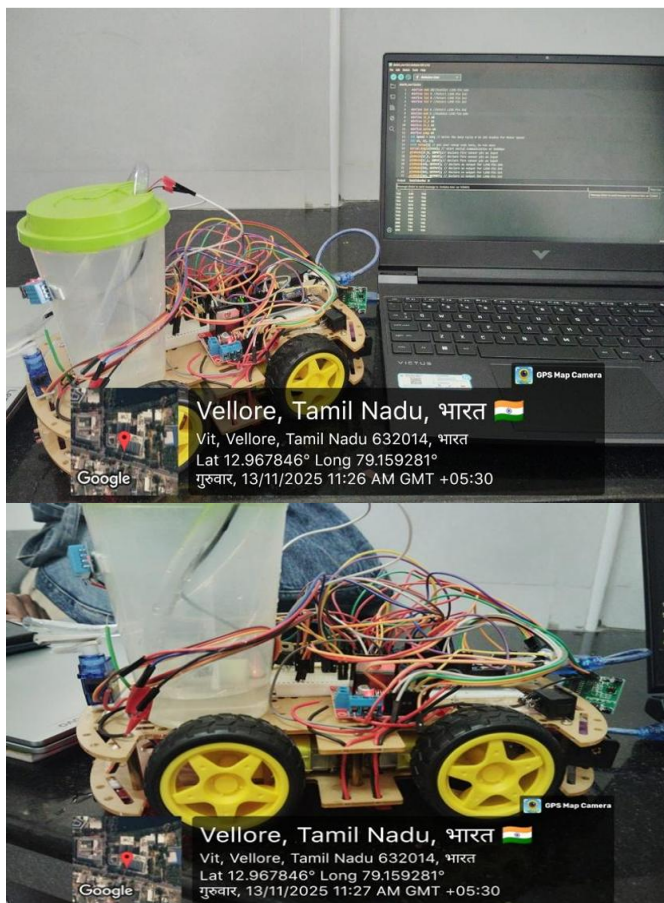
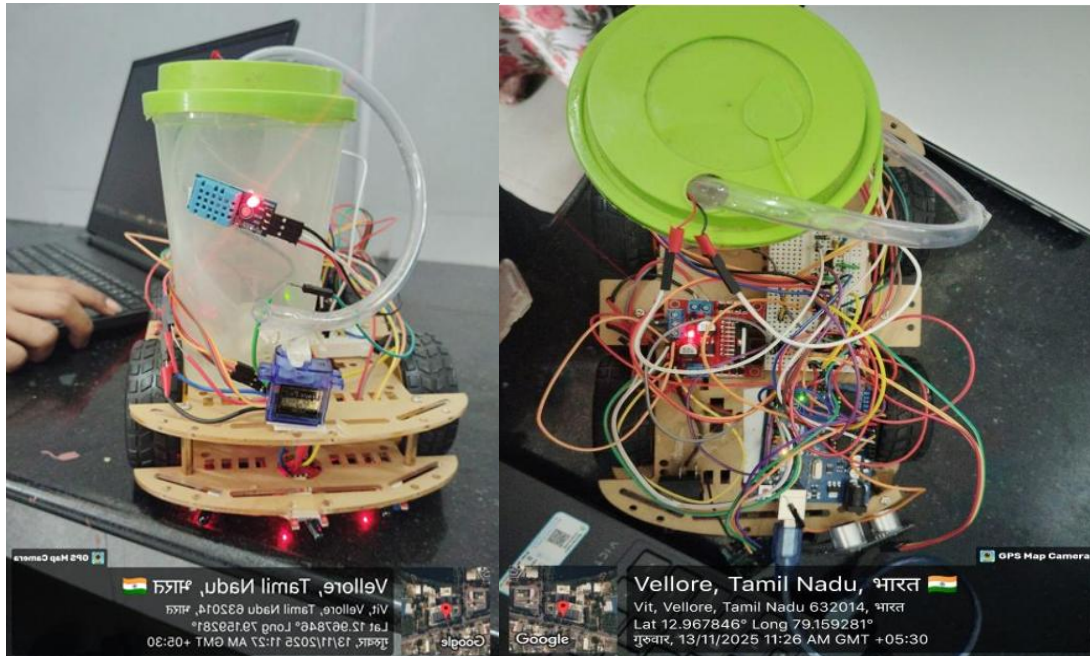
- 1. Microcontroller
 - Arduino Uno – Brain of the robot; reads sensors and controls motors & fire extinguisher. Alternatives: Raspberry Pi (vision-based), STM32 (high performance).
- 2. Sensors
 - Flame Sensor: Detects fire.
 - Ultrasonic Sensor: Obstacle avoidance.
 - Optional: IR sensors, temperature sensor, IMU.
- 3. Motors / Actuators

- DC Motors with Wheels – Movement.
- Motor Driver (L298N) – Controls motor direction & speed.
- Servo Motor – Aims fire extinguisher.
- Mini Pump / Extinguisher – Sprays water or fluid.
- 4. Power System • Battery: Li-ion or Li-Po (7.4–12V).
- Voltage Regulator: Supplies Arduino and sensors (5V).
- Ensures sufficient current for motors and actuators.

Software:

- Platform: Arduino IDE (C/C++).
- Modules:
 - 1.Initialization: Setup sensors, motors, and communication.
 - 2.Navigation: Move forward; avoid obstacles using ultrasonic sensor.
 - 3.Fire Detection: Flame sensor detects fire → stop and aim extinguisher.
 - 4.Fire Extinguishing: Servo positions nozzle, pump sprays water.
 - 5.Resume Operation: Continue navigation after extinguishing.
- Logic: Sensors → Arduino processes → Motors/Servo/Pump act.
- Optional: PID for smoother movement, LCD/buzzer alerts.

Implementation:



Testing & Results:

1. Environment Setup
 - Lab Conditions: Controlled room with small flame sources, obstacles, and flat surface.
 - Field Conditions (optional): Simulated fire scenario in safe outdoor/industrial area.
2. Performance Metrics
 - Speed: Time taken to reach fire source.
 - Accuracy: Success rate of detecting and extinguishing fire.
 - Power Consumption: Battery usage during operation.
 - Navigation Efficiency: Obstacles avoided vs collisions.
3. Results Presentation
 - Tables/Charts:
 - Time to extinguish fire at different distances.
 - Battery consumption vs operation time.
 - Obstacle avoidance success rate.
 - Graphs: Performance comparison of multiple runs or sensor setups
4. Comparison with Baseline

Method	Speed	Accuracy	Power
Manual robot	Medium	70%	High
Proposed Arduino robot	Fast	90%	Medium

•Shows improvement in autonomy, speed, and accuracy compared to manual or basic robots.

Video of the Project:

[VID-20251114-WA0045.mp4 - Google Drive](#)

Applications:

1. Real-World Use Cases
 - Industrial: Detect and extinguish fires in factories, warehouses, or chemical plants.
 - Medical/Healthcare: Fire safety in hospitals or laboratories where human intervention is risky.
 - Defense: Autonomous fire suppression in ammunition depots or hazardous military zones.

- Agriculture: Control small fires in greenhouses or crop storage areas.
- 2. Scalability & Adaptability
- 3. • Scalability: Can be upgraded for larger areas by adding more robots or advanced sensors.
- 4. • Adaptability:– Add vision-based fire detection (camera + AI).– Integrate GPS for outdoor navigation.– Modular design allows swapping sensors or extinguishing methods.

Challenges & Limitations

1. Technical Difficulties

- Accurate fire detection in presence of ambient light.
- Obstacle avoidance in cluttered environments.
- Ensuring stable navigation on uneven surfaces.
- Coordinating motors and extinguishing mechanism simultaneously.
- Limited battery life affecting operation time.

2. Constraints

- Budget: Low-cost components limit advanced sensors or actuators.
- Size: Compact chassis restricts larger pumps or heavy batteries.
- Environment: Indoor lab setup may not fully replicate real fire conditions.
- Safety: Must handle fire safely during testing without risk to surroundings.

Future Scope

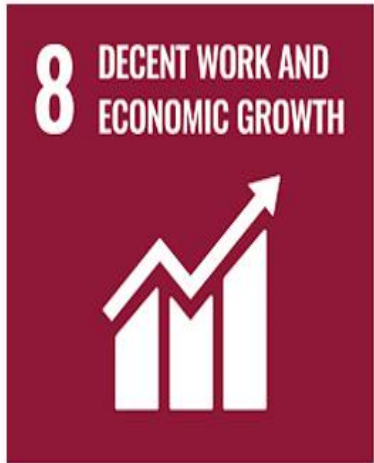
1. Planned Upgrades

- AI Integration: Use cameras with machine learning (e.g., OpenCV or TensorFlow) for advanced fire detection.
- Improved Sensors: LiDAR or thermal cameras for better navigation and fire localization.
- Enhanced Autonomy: PID control, path planning algorithms, and multi-robot coordination.
- Power Efficiency: Use higher-capacity batteries or energy-optimized components.

2. Research Extensions

- Outdoor Fire Fighting Robots: Adapt for forests or industrial outdoor setups.
- Swarm Robotics: Multiple robots working collaboratively to extinguish large fires.
- IoT Integration: Remote monitoring and control via smartphone or cloud.
- Adaptive Learning: Robot learns from past operations to improve speed and accuracy.

Sustainable Development Goals:



SDG 8 – Decent Work & Economic Growth

Fire extinguishers promote safe, hazard-free workplaces by preventing fire-related injuries and property loss. Their presence ensures compliance with safety standards, reduces downtime, and supports continuous operations. This fosters employee well-being, enhances productivity, and contributes to sustainable economic growth through secure and resilient employment environments.



SDG 9 – Industry, Innovation & Infrastructure

Integrating fire extinguishers into industrial and public infrastructure enhances safety and resilience. They are essential for risk mitigation in factories, labs, and buildings. Advanced fire suppression technologies also reflect innovation in safety systems, supporting sustainable industrialization and infrastructure that can withstand emergencies and protect both assets and lives.



SDG 11 – Sustainable Cities & Communities

Fire extinguishers contribute to urban safety by enabling quick response to fire hazards in homes, schools, and public spaces. Their accessibility reduces fire-related damage and casualties, especially in densely populated areas. This supports disaster preparedness, strengthens community resilience, and aligns with the goal of building inclusive, safe, and sustainable cities.

Conclusion

Key Takeaways

- Arduino-based autonomous fire-fighting robots can detect and extinguish fire without human intervention.
- Combining sensors, actuators, and microcontrollers enables safe and efficient navigation in hazardous environments.
- Low-cost, modular design makes it accessible for educational and practical applications.
- Performance depends on sensor accuracy, power management, and programming logic.

Impact of the Project

- Safety: Reduces human risk in fire-prone areas.
- Efficiency: Faster fire detection and suppression.
- Innovation: Demonstrates practical robotics applications in safety and automation.
- Scalability: Can be adapted for industrial, medical, defense, and agricultural use.

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